INORGANIC CUMULATIVE EXAM

December 7, 1991

The attached excerpts are from a recent article in Inorganic Chemistry. Show your understanding of this article by answering the following questions.

1) Under figure 1 is a name that involves the symbols μ_3-η^2 and μ-η^2. Explain what these symbols mean.

2) What does "dppm" (in scheme I and in scheme II and through-out the article) stand for?

3) Does the Ru atom on the right side of complex 6 obey the "18 electron rule"? Explain.

4) Complexes 1 and 3 and 6 are referred to as "48 electron" complexes. This is a special way of counting for a tri-metal system which involves most of the normal electron counting "rules" (as used, for example, in question 3 above). Explain the 48 total for 6.

5) Counting in the same way that produces 48 for 1, 3, and 6, what is the electron count for 2? For 4? For 5?

6) Give a reasonable reason for the fact that one Ru-Ru bond is longer than the other two in the structure of figure 1.

7) The IR (carbonyl region) of 6 was reported. What would you expect, in general, for this region of the IR of 6?

8) Show you follow the NMR explanation (last paragraph) for isomers 5a and 5b by explaining the circled letters - what they mean and why they are what they are.

A Highly Reactive Triruthenium Cluster Complex Bearing an Alkyne Coordinated in the μ_3-η^2-Perpendicular Mode

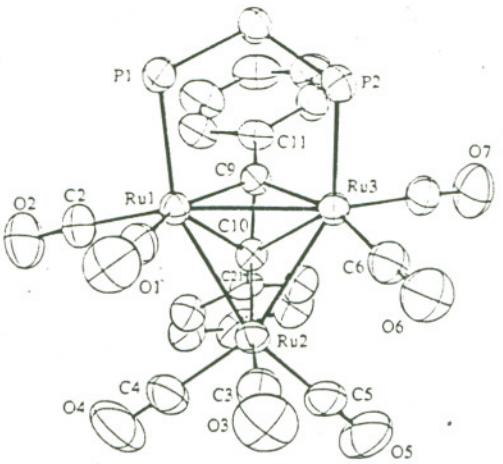
Twenty-five years after the pioneering discovery of the first ("unsaturated" alkyne) triiron nonacarbonyl complex, $Fe_3(\mu_3\text{-}\eta^2\text{-RCCR})(CO)_9$, by Dahl et al.,[1] we report the first ruthenium compound of a related type.

The unique 46-e iron complex was originally regarded as a curiosity, in that its alkyne moiety was found to lie perpendicular to one metal–metal bond, in sharp contrast to the majority of known ruthenium or osmium complexes that are saturated 48-e species exhibiting a parallel orientation of the alkyne with respect to a metal–metal bond.[2] , we were prompted to investigate the effect of phosphine ligands on the anionic species $[Ru_3(\mu\text{-Cl})(\mu_3\text{-}\eta^2\text{-RCCR})(CO)_9]^-$, (1) which is readily available via a halide-promoted reaction of alkynes with $Ru_3(CO)_{12}$.[8]

The reaction of the PPN salt of (1) (PPN = bis(triphenylphosphine)nitrogen cation) with dppm was found to proceed spontaneously in THF solution at 25 °C with concomitant precipitation of (PPN)Cl. The stoichiometric reaction between PPN(1) and 1 equiv of dppm led to a mixture of two neutral complexes, namely a green compound **2** (a very unusual color for a ruthenium carbonyl cluster) and a violet species, **3**, in approximate ratio 1/10 (Scheme I, path i).

The isolated green species could be also cleanly converted into the violet one in contact with CO at 25 °C within about 10 s! The reverse transformation leading to quantitative yields of the green species was achieved within 1 h in refluxing THF under a stream of inert gas.[10] We rapidly realized that we were just observing for the first time the clean and reversible interconversion between an unsaturated alkyne–triruthenium cluster and its CO adduct. The green species was subsequently crystallized and formulated as the [redacted]-e complex $Ru_3(\mu_3\text{-}\eta^2\text{-}(C_6H_5)CC(C_6H_5))(\mu\text{-}\eta^2\text{-dppm})(CO)_7$ (**2**), analogous to the above-mentioned osmium derivative.[7] Its molecular structure is shown in Figure 1.[11] Unlike its osmium analogue, the ruthenium complex is reactive toward CO (Scheme I), [redacted]osphines, and hydrogen (Scheme II), in all cases at 25 °C. The violet CO adduct was formulated as $Ru_3(\mu_3\text{-}\eta^2\text{-}(C_6H_5)CC(C_6H_5))(\mu\text{-}\eta^2\text{-dppm})(\mu\text{-CO})(CO)_7$ (**3**).[10,12] The reaction of **2** with dppm at 25 °C was found to produce the $(dppm)_2$ derivative $Ru_3(\mu_3\text{-}\eta^2\text{-}(C_6H_5)CC(C_6H_5))(\mu\text{-}\eta^2\text{-dppm})_2(CO)_6$ (**4**) (yield 53%),[13] Finally, the reaction of **2** with hydrogen (1 atm, 10 min, 25 °C) provided the dihydrido complex $Ru_3(\mu\text{-H})_2(\mu_3\text{-}\eta^2\text{-}(C_6H_5)CC(C_6H_5))(\mu\text{-}\eta^2\text{-dppm})(CO)_7$ (**5**) in 89% yield.[15] An NMR study of the solution at −80 °C revealed the occurrence of two isomers in solution (approximate ratio **5a**/**5b** = 10/1). The X-ray structure of **5a** was determined.[16]

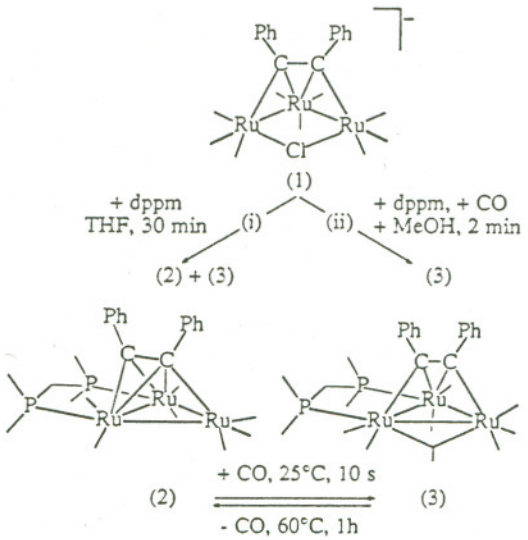
In parallel to these observations, we attempted to determine whether the presence of one monodentate phosphine ligand (instead of dppm) on the alkyne–triruthenium framework would be sufficient to stabilize a perpendicular orientation of the alkyne. The answer proved to be negative for triphenylphosphine whose reaction with PPN(**1**) gave only the saturated 48-e complex $Ru_3(\mu_3\text{-}\eta^2\text{-}(C_6H_5)CC(C_6H_5))(\mu\text{-CO})(CO)_8(PPh_3)$ (**6**),[17,18] with no evidence for a corresponding unsaturated species.



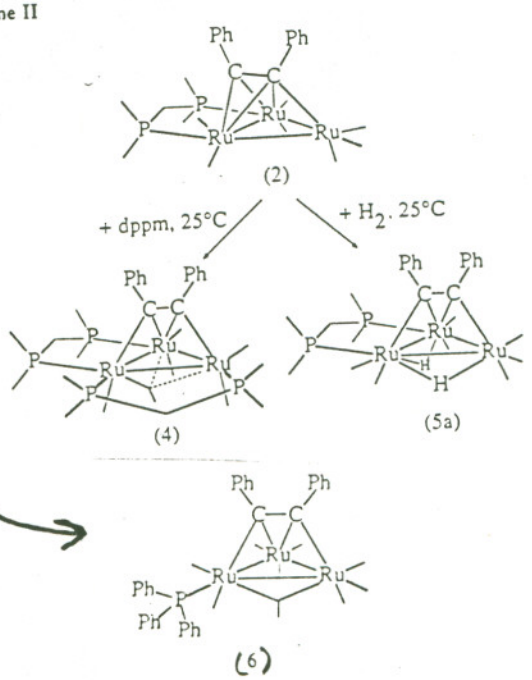
(2)

Figure 1. Perspective view of the molecular unit of the complex $Ru_3(\mu_3\text{-}\eta^2\text{-}(C_6H_5)CC(C_6H_5))(\mu\text{-}\eta^2\text{-}([redacted])(CO)_7$ (**2**). The [redacted] have been omitted for clarity. Selected interatomic distances (Å) with esd's in parentheses: Ru(1)–Ru(2) = 2.6520 (6), Ru(2)–Ru(3) = 2.6683 (6), Ru(1)–Ru(3) = 2.8120 (7), Ru(1)–P(1) = 2.317 (1), Ru(3)–P(2) = 2.298 (1), Ru(1)–C(9) = 2.108 (4), Ru(3)–C(9) = 2.083 (4), Ru(1)–C(10) = 2.202 (4), Ru(3)–C(10) = 2.265 (4), Ru(2)–C(10) = 2.202 (4), C(9)–C(10) = 1.409 (6).

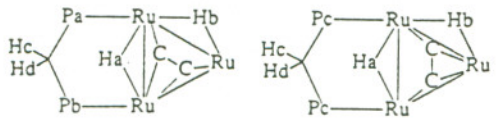
Scheme I



Scheme II



Two isomers were detected by NMR at −80 °C; they differ in the orientation of the alkyne with respect to the edge-bridging dppm ligand. P and H nuclei are labeled as indicated. Isomer **5a**: ^{31}P NMR



Isomer 5a Isomer 5b

(200 MHz, $CDCl_3$) δ 17.51 (d, P_a, $J(P_a\text{–}P_b)$ = 51 Hz), 8.38 (d, P_b, $J(P_a\text{–}P_b)$ = 51 Hz). ^1H NMR (200 MHz, $CDCl_3$) δ 7.5–6.4 (m, 30 H, C_6H_5), 3.04 (m, 2 H, CH_2, $J(H_c\text{–}P_b) = J(H_d\text{–}P_b)$ = 13 Hz, $J(H_c\text{–}H_d)$ = 11 Hz, $J(H_c\text{–}P_a) = J(H_d\text{–}P_a)$ = 12 Hz), −14.01 (dd, H_a (μ-hydride), $J(P_a\text{–}H_a)$ = 14 Hz, $J(P_b\text{–}H_a)$ = 12 Hz), −18.89 (t, H_b (μ-hydride), $J(P_a\text{–}H_b)$ = 37 Hz). Isomer **5b**: (tentative assignment) ^{31}P NMR (200 MHz, $CDCl_3$) δ 39.94 (s, 2 P_c). ^1H NMR (200 MHz, $CDCl_3$) δ 7.5–6.4 (m, 30 H, C_6H_5), 2.27 (dd, H_c, CH_2, $J(H_c\text{–}H_d)$ = 13 Hz, $J(H_c\text{–}P_c)$ = 21 Hz), 3.51 (dd, H_d, CH_2, $J(H_c\text{–}H_d)$ = 13 Hz, $J(H_d\text{–}P_c)$ = 22 Hz), −15.25 (s, H_a (μ-hydride), $J(P_c\text{–}H_a)$ = 5 Hz), −18.76 (t, H_b (μ-hydride), $J(P_c\text{–}H_b)$ = 15 Hz).

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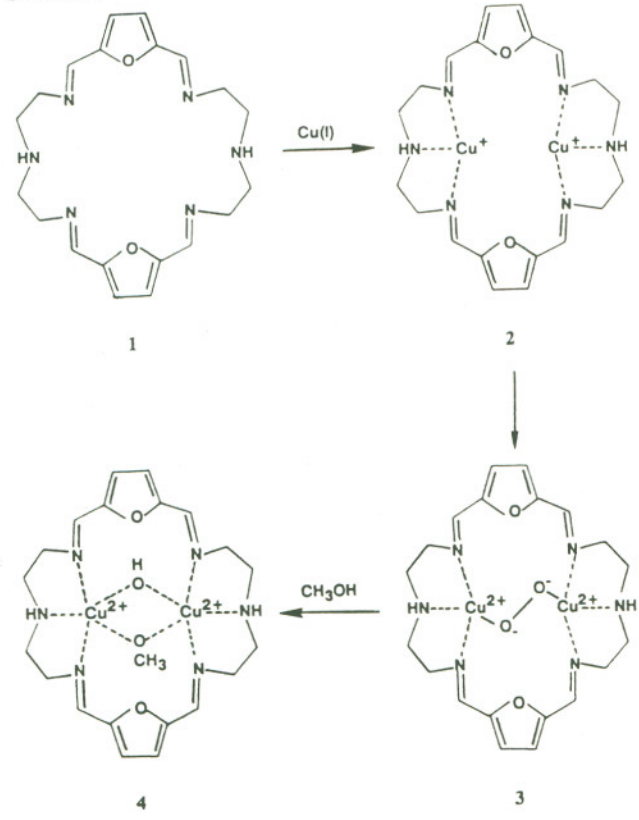
1) The following figures appeared in papers in recent additions of **Inorganic Chemistry** . Briefly discuss any interesting or unique features of the compound under study and suggest a reason for the interest in this research. Choose 4 examples to discuss.

1)

Scheme I



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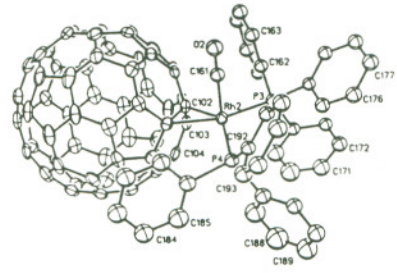

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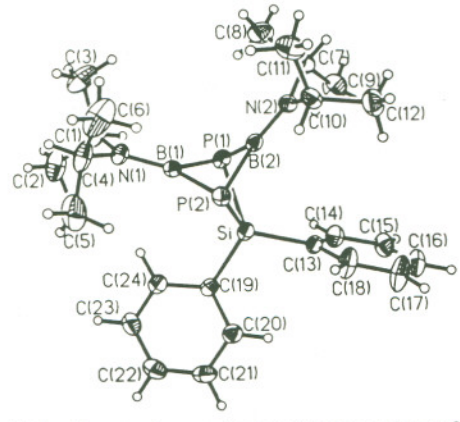


Figure 1. Molecular structure and atom labeling scheme for

4)

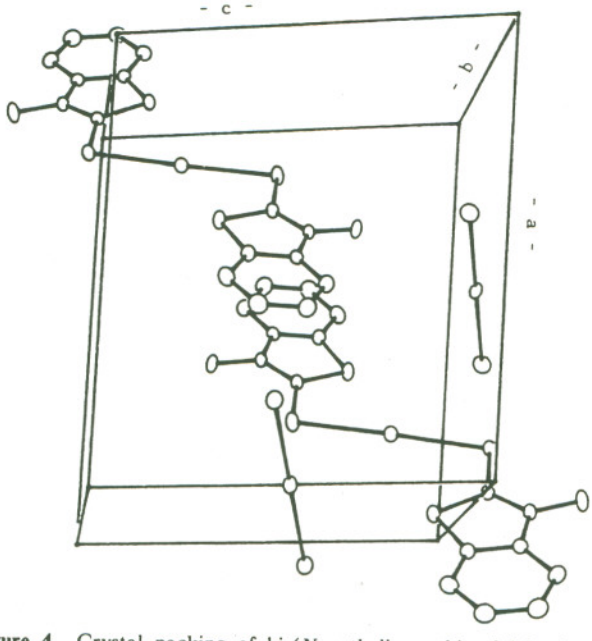
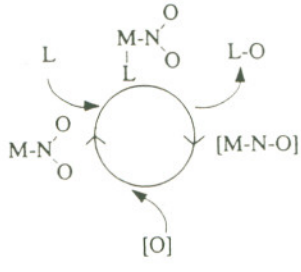


Figure 4. Crystal packing of bis(*N*-methylbenzothiazolyl-2-seleno)-iodonium triiodide. The unit-cell contains two centrosymmetrically related $[(2)_2I]^+$ cations in a general position of the unit cell and two I_3^- located on two different crystallographic inversion centers.

INVERSION CENTERS

5)

M = CpCr



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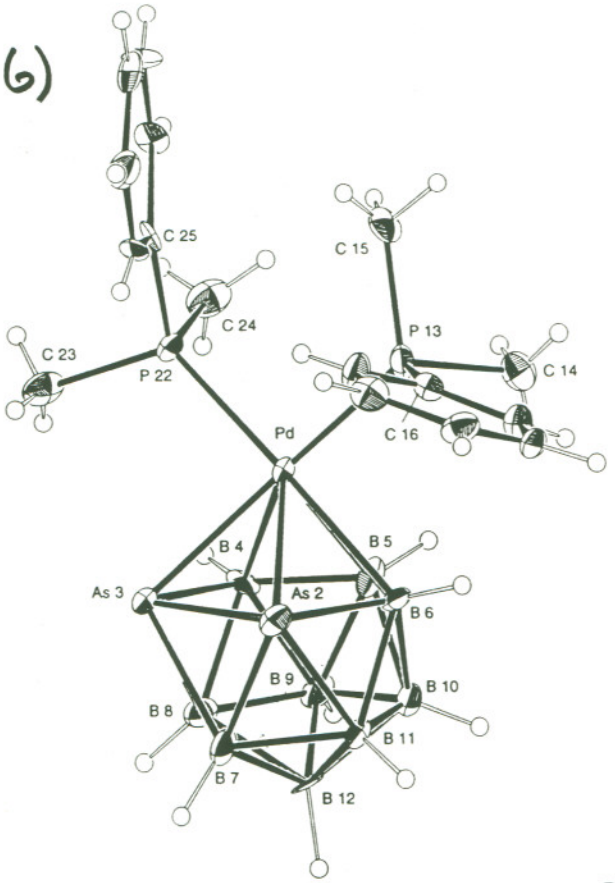
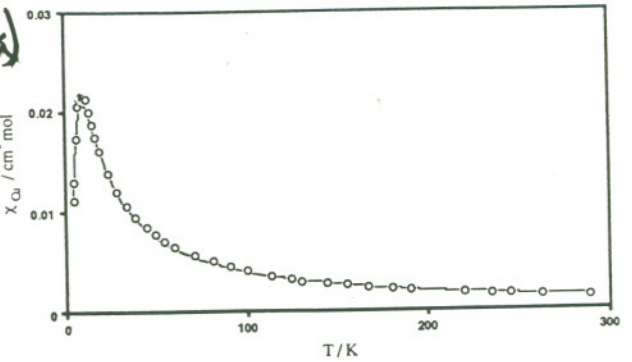


Figure 2. ORTEP diagram of 1,1-$(PMe_2Ph)_2$-*closo*-1,2,3-$PdAs_2B_9H_9$ (3).

7a)



7b)

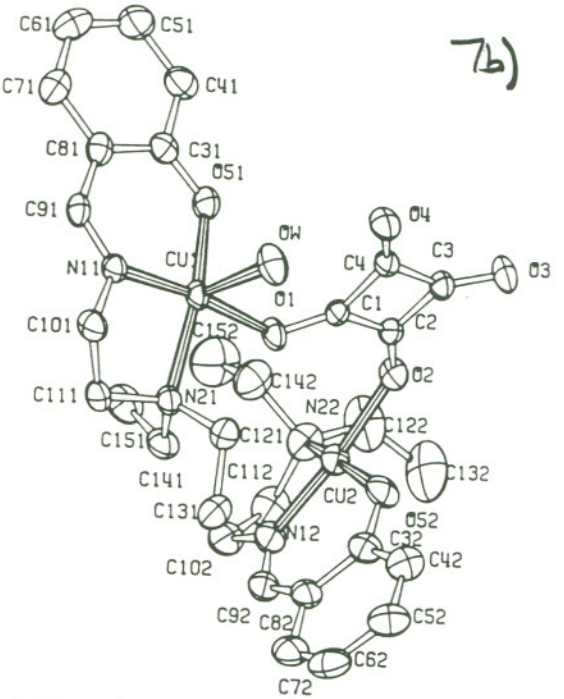
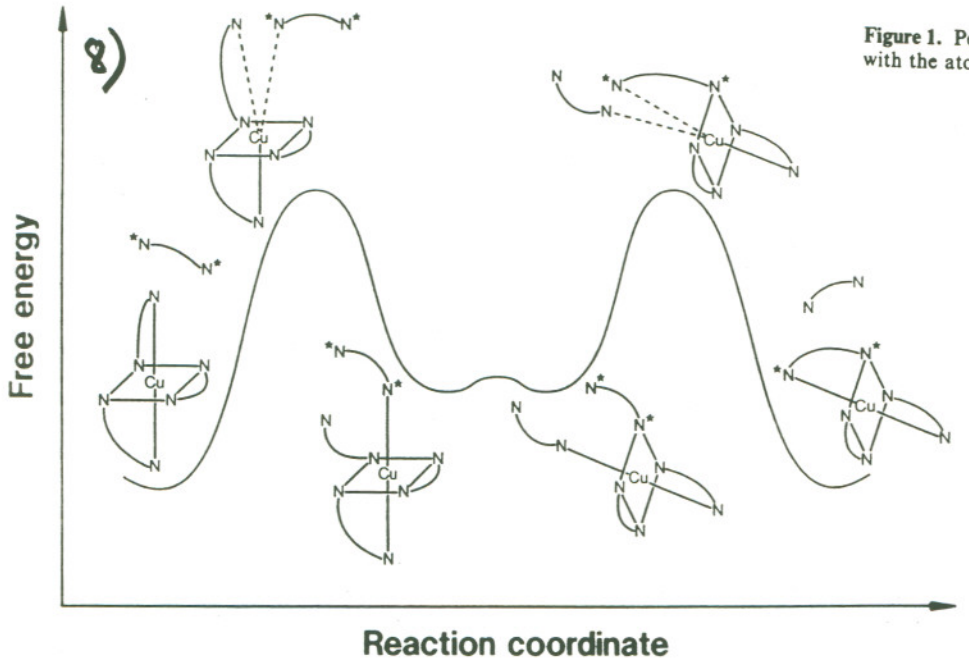


Figure 1. Perspective view of $[Cu_2(SalNEt_2)_2(H_2O)(C_4O_4)]\cdot H_2O$ along with the atom-labeling scheme.

8)



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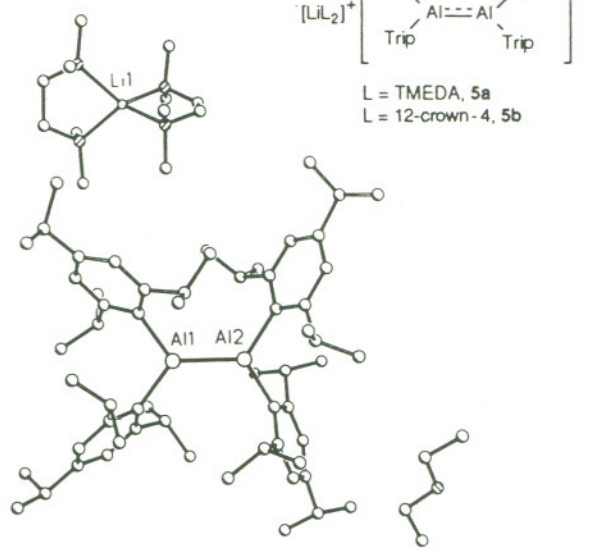
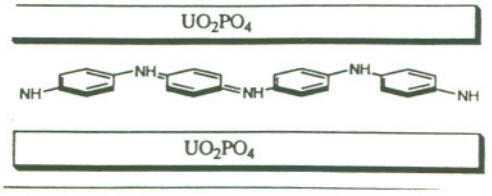


10)



hint: aniline